

Activity Guide — middle / module-11-multiplication-near-base

IKS Curriculum

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Activity 01 — Mixed-Sign Drill

Module: Nikhilam Multiplication · **Band:** Middle · **Day:** 5 · **Time:** 20 min

Type

Paired practice with self-correction.

Goal

Each student solves 8 mixed-sign Nikhilam problems (one factor above 100, one below), with the partner checking accuracy.

Materials per pair

- 1 printed problem sheet (8 problems below).
- 1 pen each.
- 1 stopwatch (optional — for self-pacing).

Set-up (teacher, before class)

- Print the problem sheet.
- Have the answer key on a separate slip for the pair to self-check after they finish.

Instructions (student-facing handout)

Mixed-Sign Drill — work in pairs.

1. Solo (8 min): each partner solves all 8 problems individually. Don't peek.
2. Pair check (7 min): swap sheets. Partner A reads problem 1 aloud; partner B resolves it from scratch on a fresh strip. Compare answers. If they differ, work out which is right.
3. Last 5 min: each student redoes any 3 problems they got wrong, silently.

Problem sheet

- | | |
|--------------------|--------------------|
| 1. 102×98 | 5. 97×104 |
| 2. 104×97 | 6. 105×95 |
| 3. 103×96 | 7. 101×99 |
| 4. 106×95 | 8. 108×92 |

Answer key (teacher copy — DON'T print on student sheet)

Q	Signed parts	Left	Right (signed)	Answer
1	+2, -2	100	-4	9996
2	+4, -3	101	-12	10088
3	+3, -4	99	-12	9888
4	+6, -5	101	-30	10070
5	-3, +4	101	-12	10088
6	+5, -5	100	-25	9975
7	+1, -1	100	-1	9999
8	+8, -8	100	-64	9936

Discussion prompts (last 2 min)

- Which problem took the longest? Why?
- Did you notice that Q7 (101×99) is exactly $(100+1)(100-1) = 100^2 - 1^2 = 9999$? That's the difference-of-squares identity. Nikhila IS that identity for the special case.
- When excess equals deficit (Q1, Q6, Q7, Q8), the left is always exactly the base. Did you spot this shortcut?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Partners writing deficits clearly, debating sign carefully	"Wait, $+4 \times -3$ is -12 , not $+12$."
Needs intervention	One partner just copies the other; or both ignore signs	"I got 10100, you got 10088 — which is it?" → great teachable moment

Differentiation

- **Tier up:** Try 121×79 (large mixed gap; difference of squares — answer 9559). Try 1003×997 (mixed, near 1000 — answer 999991).
- **Tier down:** Stay with 102×99 , 103×98 , 101×99 (small differences) until the sign care is automatic.

Safety / sensitivity

- No timed competition between pairs publicly. Pairs are self-pacing.

Clean-up

- Worksheets collected (for the teacher to spot-check before Day 6).

Activity 1 — Method Speed Race

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Middle · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of Multiplication Near a Power of 10 — Nikhilam through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

20 problem cards, stopwatches, two columns on whiteboard

What students do

Split class in two: one team solves with traditional long multiplication, the other with Nikhilam. Race on identical problems. Discuss which problems favour which method.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 02 — Speed Sprint

Module: Nikhilam Multiplication · **Band:** Middle · **Day:** 9 · **Time:** 18 min within Day 9

Type

Timed individual drill + paired chorus round.

Goal

Each student solves 10 mixed-case Nikhilar problems mentally in 3 minutes; then pairs do a chorus-round of 10 more.

Materials per pair

- 1 sprint sheet (10 problems).
- 1 pen each.
- 1 stopwatch (or use the wall clock).

The sprint problem sheet (round 1)

- | | |
|---------------------|------------------------|
| 1. 97×98 | 6. 88×89 |
| 2. 103×104 | 7. 102×99 |
| 3. 104×105 | 8. 99×99 |
| 4. 105×95 | 9. 998×997 |
| 5. 96×96 | 10. 1004×1003 |

The sprint problem sheet (round 2, alternate)

- | | |
|---------------------|------------------------|
| 1. 95×94 | 6. 89×92 |
| 2. 107×106 | 7. 103×96 |
| 3. 108×104 | 8. 101×101 |
| 4. 106×94 | 9. 997×998 |
| 5. 93×97 | 10. 1005×1004 |

Instructions

Speed Sprint — individual, 3 minutes per round.

1. **Round 1 (3 min):** Solve all 10 problems on sheet 1 as fast as possible. Mental, not paper-and-pencil if you can.
2. **Self-correct (1 min):** Use the answer key.
3. **Round 2 (3 min):** Solve all 10 on sheet 2.
4. **Self-correct (1 min).**
5. **Chorus round (5 min):** Pair up. Partner A reads problem; you both shout the answer. Switch. Cover 10 random problems.

Round 1 answers

Q	Answer
1	9506
2	10712
3	10920
4	9975
5	9216
6	7832
7	10098

Q	Answer
8	9801
9	995006
10	1007012

Round 2 answers

Q	Answer
1	8930
2	11342
3	11232
4	9964
5	9021
6	8188
7	9888
8	10201
9	995006
10	1009020

Scoring

- 10 correct in 3 min = “Exceeds” (sub-20-sec average).
- 7–9 correct = “Meets.”
- 4–6 correct = “Approaches” — practise more before Day 10’s quiz.
- 0–3 correct = “Beginning” — get a teacher 5-min reteach on Day 10.

Going well / Needs intervention

	Looks like	Sounds like
Going well	Quiet sustained focus, fast pencil-strokes, deficit-then-answer	(Nothing — they’re solving)
Stuck	Confused face, restarting same problem 3 times	“Wait, is 88 above or below 100?”

Differentiation

- **Tier up:** Sprint round of 20 problems in 3 minutes; or include 4-digit problems near 10000.
- **Tier down:** 5 problems in 3 minutes (gentler pace); focus on accuracy over speed.

Clean-up

- Sheet collected (for retest record).
- Pair-chorus round is unscored — just for chorus energy.

Project Brief

Project Brief — Module 11 Nikhilam Multiplication (Middle)

Work sessions: Day 9 (planning) + home work · **Presentation:** Day 10 · **Total marks:** 40 (project 30 + presentation 10)

The project

Each pair (or individual, with teacher approval) chooses ONE of three formats:

Format 1: 60-second teaching video

Make a short video that teaches Nikhilam multiplication to a Grade 4 or 5 student. Constraints: - Total length: 60 seconds (± 10 sec). - Must show: **at least one** worked example clearly (deficits, cross-subtract, right-part with pad, final answer). - Bonus: include both a both-below-100 example AND a both-above-100 example. - Cite the source (PDF + sūtra name) in the title slide or closing credit.

Format 2: Printed workbook

Produce a 20-problem workbook for a Grade 4 or 5 self-learner with Nikhilam multiplication. Constraints: - 20 problems organised in 4 difficulty levels (5 problems each). - Level 1: single-digit $\times$ single-digit (base 10). - Level 2: both below 100 (no overflow). - Level 3: both below or both above 100 (mix). - Level 4: mixed-sign + near-1000. - Each level includes ONE fully worked example. - Answer key on the back page. - Title page with cited source (PDF + sūtra name). - Optional: a 1-page “when NOT to use this” note (from Day 8).

Format 3: Comparative speed-test data project

Time 5–10 people (family, schoolmates) on a small set of multiplication problems, comparing long multiplication vs. Nikhilam. Constraints: - 4–6 problems: a mix of Nikhilam-suitable (e.g. 97×96 , 103×104 , 998×997) and Nikhilam-unsuitable (e.g. 47×53). - A table of names (or codes), problem solved, time using standard, time using Nikhilam (if applicable). - A plot or visual comparison. - A 1-paragraph interpretation of the data — including any surprising findings (e.g. “*Nikhilam was only faster when both numbers were near 100; for 47×53 it was slower*”). - Cite the source PDF.

Why this matters

You’re moving from *learner* to *teacher* (or *researcher*). The discipline of explaining or measuring forces you to understand more deeply.

You’ll also learn the meta-lesson: the Nikhilam method is fast in its niche and slow outside it. A good project shows both.

What success looks like

A strong project will: - Demonstrate the Nikhilam method accurately (no arithmetic errors). - Be pitched for the intended audience (a 9-year-old for video and workbook; a peer reader for data). - Show the pair's own thinking. - Be clearly communicated. - Acknowledge the *limit* of the method (when it's faster, when it's not).

Working in pairs

- Pair work strongly encouraged.
- Individual allowed for the workbook or video format.
- Groups of 3 acceptable — each member must have a distinct role.

Citation rules

- At minimum **one** citation must appear in your project — visible.
- Format: <PDF filename> (e.g. *Vedic Mathematics Batch1.pdf*).
- Naming the sūtra (*Nikhilam Navataścaramaṁ Daśataḥ*) is encouraged.
- See `sources.md` for the list of available PDFs.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to learn or make, and what did you actually end up with? 2. What's the most surprising thing you discovered while making this? 3. If you had three more days, what would you do differently?